

Mohan Konugolu

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I am a seasoned robotics engineer with over 4 years of hands-on experience developing and deploying mobile robots across diverse industries, which helped me gain essential skills like teamwork, adaptability and damage control. My expertise lies in rapid prototyping and efficient deployment of robotic solutions.

PROFESSIONAL EXPERIENCE

Robotics Freelancer - Upwork

Feb 2024 - Present

- Setting up simulation (isaacsim), training robots using reinforcement learning for specific tasks (isaacsim) and porting entire software stacks to ros2 and isaac sim.

Software Developer - Bosch (BGSW)

Sep 2022 - Feb 2024

- Implemented ML algorithms for autonomous driving, specializing in computer vision for depth perception and reinforcement learning for vehicle control.

Robotics Engineer - ZeroPoint Robotics

Aug 2021 - 30 Aug 2022

- Developed a complete software stack for an autonomous material handling mobile robot.

Deployment Engineer - THRSL

Dec 2020 - Jul 2021

- Developed perception code for pallet detection and executed robot deployment.

EDUCATION

MSc. Robotics and Artificial Intelligence

Cardiff Metropolitan University

Sep 2024 - Sep 2025

Cardiff, United Kingdom

BEng. Mechatronics

Chandigarh University

Jul 2016 - Jul 2020

Punjab, India

PROJECTS

Custom Humanoid simulation and RL locomotion

Imported an open source humanoid by hyperspawn with its complex 4-bar mechanisms and stuart platform and then trained the robot for locomotion in isaac sim using isaacsim.

- Solved the complex kinematic linkage of 4-bar mechanisms and stuart platform in isaac sim and made sure everything works just like the real robot.

- Trained the humanoid to walk using reinforcement learning and integrated it with a motion retargeting pipeline for SMPLX (PHC) to train using imitation and reinforcement learning.
- Mentioned as one of the contributors in the hyperspawn open source project. [github link](#)

Brick Laying Using Reinforcement Learning

Trained a simulation of an industrial arm with 5 fingered humanoid hand to pick and place bricks using reinforcement learning.

- Created a virtual environment in Isaac sim with a custom robot and integrated it with Isaac lab library for reinforcement pipeline. Added force sensors to the tips of all 5 fingers in the hand along with IMU sensors.
- Used the SKRL library and PPO algorithm to train the robotic arm.
- Trained the network to slowly approach the brick with an open arm, grasp the brick with all 5 fingers, lift and take the brick to the desired location.

Depth Estimation Using Mono-Camera

Generated absolute object level depth from a monocular camera by training a deep learning model. Object bounding boxes (YOLO) and absolute depth image (MiDas algorithm) were used as network input and depth of the detected object's centroid as output.

- When tested on our dataset created on our in-house test vehicle a MAE of 0.18m was achieved whereas on nuScenes dataset with a lot of different scenes a MAE of 1.1m was achieved.
- For this project we were invited for a poster presentation in Bosch Germany in the AAI conference held for internal bosch A.I research.

Autonomous Mobile Robot Software stack

Being one of the core software engineers in the startup, I have worked on the complete robot stack from safety code to motion planner code.

- Developed a double look ahead path tracking algorithm based on pure pursuit.
- Implemented double continuous curve algorithms using clothoid for smooth and constraint-based path generation.
- Implemented industry-ready TEB for more smooth yet computationally inexpensive local paths.
- Coded laser-based safety with 3 zones using 2D lidar.
- Developed a trolley leg detection algorithm using 2D lidar.

Note: My [upwork](#) profile showcases the recent freelancing projects i have done, mainly on isaac sim and reinforcement learning for many robotics companies while pursuing my masters!

SKILLS

Deep Learning / Reinforcement Learning / PyTorch / Git / OpenCV / Python / C++ / Linux / ROS / ROS2 / docker / NVIDIA Isaac Sim / Gazebo